Exact Local Obstructions for Two Quadratic Prime Values

A rigorous sieve-theoretic partial result for OEIS A232768

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Abstract

OEIS A232768 records the positive integers n for which both $F(n) = 2n^2 + 2n + 1$ and $G(n) = 3n^2 + 6n + 5$ are prime. Whether there are infinitely many such n remains open. We calculate exactly, for every prime p, the number of residue classes for which $p \mid F(n)G(n)$. This proves admissibility of the polynomial pair and yields an explicit positive density of integers avoiding all obstructions from *any specified finite set of primes*. The result is local and does not prove infinitely many simultaneous prime values.

The precise OEIS problem and scope

The entry A232768 asks for the positive n such that

$$F(n) = n^2 + (n+1)^2 = 2n^2 + 2n + 1, \qquad G(n) = n^2 + (n+1)^2 + (n+2)^2 = 3n^2 + 6n + 5$$

are simultaneously prime. Its comments identify infinitude of prime values of either quadratic as unresolved [1]. In particular, the stronger simultaneous-primality infinitude question is unresolved. The result below proves the *local admissibility* needed for prime-value heuristics; it is neither a proof of the infinitude conjecture nor a claim of mathematical novelty.

For a prime p, put

$$\nu(p) := \#\{r \pmod p : F(r)G(r) \equiv 0 \pmod p\}.$$

For odd p, let $\left(\frac{a}{p}\right)$ be the Legendre symbol.

Theorem 1 (Exact local obstructions)**.** *For the polynomial pair defining A232768,*

$$\boxed{\nu(p) = \begin{cases} 1, & p = 2, \\ 0, & p = 3, \\ 3, & p = 5, \\ 2 + \left(\frac{-1}{p}\right) + \left(\frac{-6}{p}\right), & p \geq 7. \end{cases}}$$

In particular, $\nu(p) < p$ for every *prime p. Equivalently, no prime divides $F(n)G(n)$ for all integers n.*

Proof. Modulo 2, $F(r) \equiv 1$ and $G(r) \equiv r + 1$; precisely the odd residue is forbidden. Modulo 3, $G(r) \equiv 2$ and $F(0), F(1), F(2) \equiv 1, 2, 1$, so $\nu(3) = 0$.

For every odd prime p, the identity

$$2F(r) = (2r+1)^2 + 1$$

shows that $F(r) \equiv 0 \pmod p$ has $1 + \left(\frac{-1}{p}\right)$ roots: the affine map $r \mapsto 2r + 1$ is bijective modulo p. For $p \neq 3$ we also have

$$3G(r) = (3r+3)^2 + 6,$$

so $G(r) \equiv 0 \pmod p$ has $1 + \left(\frac{-6}{p}\right)$ roots.

It remains to count common roots. An exact integer identity gives

$$2G(r) - 3F(r) = 6r + 7.$$

For $p \geq 5$, a common root must satisfy $r \equiv -7/6 \pmod p$; substitution into F gives

$$F(-7/6) = \frac{25}{18}.$$

Since 18 is invertible for $p \geq 5$, a common root is possible only for $p = 5$. Indeed modulo 5 the F-roots are $\{1, 3\}$ and the G-roots are $\{0, 3\}$, so their union has exactly three members. For $p \geq 7$ the root sets are disjoint, establishing the displayed Legendre-symbol formula.
Finally $\nu(2) = 1 < 2$, $\nu(3) = 0 < 3$, $\nu(5) = 3 < 5$; for $p \geq 7$, $\nu(p) \leq 4 < p$. Thus the pair is admissible. □

Corollary 1 (Exact finite-prime sieve density)**.** *Let S be any finite set of primes, and put*

$$M_S := \prod_{p \in S} p, \qquad \delta_S := \prod_{p \in S} \left(1 - \frac{\nu(p)}{p}\right) > 0.$$

Then exactly $M_S \delta_S = \prod_{p \in S}(p - \nu(p))$ residue classes modulo M_S satisfy

$$\gcd(F(n)G(n), M_S) = 1.$$

Consequently, for every fixed finite S,

$$\#\{1 \leq n \leq X : \gcd(F(n)G(n), M_S) = 1\} = \delta_S X + O(M_S).$$

In particular, infinitely many positive integers evade the prime divisors in each specified finite S.

Proof. For each $p \in S$, exactly $p - \nu(p) > 0$ residues modulo p avoid both zeros. The Chinese remainder theorem puts these choices in bijection with admissible residue classes modulo the squarefree product M_S. Each full block of M_S consecutive integers has exactly $M_S \delta_S$ survivors; the remaining partial block contributes an error bounded by M_S. □

Consistency checks and significance

For $S = \{2, 3, 5, 7, 11\}$, the forbidden counts are $1, 0, 3, 2, 2$ respectively. Hence there are

$$1 \cdot 3 \cdot 2 \cdot 5 \cdot 9 = 270 \quad \text{admissible classes modulo} \quad 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310,$$

with density $270/2310 = 9/77$. The value $n = 2$ is in A232768: $F(2) = 13$, $G(2) = 29$. Also, any n giving both primes is even, since $G(n)$ is even and greater than 2 for odd $n \geq 1$. For even n, both polynomials are $1 \pmod 4$, as the OEIS comments state.
Admissibility removes a *fixed congruence obstruction*, not the global obstacle to simultaneous prime values. As S grows, M_S changes, and the above positive densities do not give a uniform lower bound on simultaneous-prime solutions. The Bateman–Horn conjecture provides a heuristic framework for such polynomial tuples [2, 3]; it is *not* used as a theorem here.

References

[1] OEIS Foundation, *A232768: Numbers n such that $n^2 + (n+1)^2$ and $n^2 + (n+1)^2 + (n+2)^2$ are both prime*, https://oeis.org/A232768, accessed November 13, 2026.

[2] P. T. Bateman and R. A. Horn, "A heuristic asymptotic formula concerning the distribution of prime numbers," *Mathematics of Computation* **16** (1962), 363–367. https://doi.org/10.2307/2004056.

[3] S. R. Garcia, "What is the Bateman–Horn conjecture?" *Notices of the American Mathematical Society* (2024), https://doi.org/10.1090/noti3046.

Verification and priority note. This document was authored by ChatGPT (OpenAI). The algebraic identities, root counts, and finite sieve examples were independently checked computationally during preparation, but no human peer review or comprehensive novelty search is claimed. This is a fully proved, OEIS-specific local result, *not* a proof of the unresolved infinitude conjecture.